# Smart Hybrid Solar-Wind Energy System with Tracking and Storage

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# DAYANANDA SAGAR COLLEGE OF ENGINEERING

(An Autonomous Institute affiliated to Visvesvaraya Technological University (VTU), Belagavi, Approved by AICTE and UGC, Accredited by NAAC with 'A' grade & ISO 9001-2015 Certified Institution) Shavige Malleshwara Hills, Kumaraswamy Layout, Bengaluru - 560 111. India

**Department of Electrical and Electronics Engineering**

(Accredited by NBA Tier 1: 2022-2025)

**CO-PO Mapping**

**COURSE OUTCOMES**

CO1: Identify a research problem by doing literature survey with supervisor's advice.

CO2: Apply the fundamental knowledge of mathematics, science and engineering principles in design of solutions of system components.

CO3: Select and apply a suitable engineering/IT tool in modeling/data interpretation /analytical studies, conduct experiments leading to a logical solution.

CO4: Design a system/ system component/process, build it and test its functioning as a solution to a complex engineering problem with sustainable practice.

CO5: Communicate effectively to a diverse audience about the outcome and societal impact; develop technical reports and publication.

**CO-PO mapping:**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 | PSO3 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | | 2 | | | | | | | | | | | 2 | 2 | 1 |
| CO2 | 1 | | | | | | | | | | | | 2 | 2 | 1 |
| CO3 | 2 | | | | 2 | | | | | | | | 2 | 2 | 1 |
| CO4 | | 3 | 3 | 3 | 3 | 3 | 3 | | | | | | 2 | 2 | 1 |
| CO5 | | | | | | 3 | | 3 | | 3 | | 3 | 2 | 2 | 1 |

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 | PSO3 |
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| CO1 | | 2 | | | | | | | | | 2 | 2 | 1 |
| CO2 | 1 | | | | | | | | | | 2 | 2 | 1 |
| CO3 | 2 | | | | 2 | | | | | | 2 | 2 | 1 |
| CO4 | | 3 | 3 | 3 | 3 | 3 | | | | | 2 | 2 | 1 |
| CO5 | | | | | | 3 | 3 | | 3 | 3 | 2 | 2 | 1 |

# ABSTRACT

This project focuses on integration of renewable energy sources has become essential to address the increasing global energy demand and mitigate environmental challenges. This project focuses on a dual power generation system combining solar and wind energy, designed to enhance efficiency and ensure a continuous energy supply. The proposed system employs photovoltaic panels to harness solar energy and a wind turbine to capture wind energy, enabling power generation even under variable weather conditions. This hybrid approach is particularly effective in regions with fluctuating solar radiation and wind patterns, maximizing energy output throughout the day and year. A shared inverter and energy storage system optimizes the cost and efficiency of the combined setup. Additionally, the design incorporates a control unit to prioritize energy sources based on availability, ensuring consistent power delivery to the load or grid. By leveraging both renewable resources, this system addresses the limitations of standalone solar or wind systems, such as intermittency and low reliability during non- ideal weather conditions. The modular design facilitates scalability for applications ranging from residential use to industrial power generation. This project demonstrates a sustainable, cost-effective solution for decentralized energy generation, promoting the transition to clean energy sources. Through simulation and real-world implementation, the dual solar-wind system showcases its potential to meet growing energy demands while reducing dependency on fossil fuels and lowering carbon emissions.

# CHAPTER 1:

## INTRODUCTION

### 1.1 BACKGROUND

The idea behind the Dual Power Generation Windmill Plus Solar Generator comes from the need for reliable, clean energy in all conditions. Solar panels work best on sunny days, while wind turbines can generate power day and night. By combining both, we can create a steady energy supply no matter the weather. The increasing demand for sustainable and reliable energy solutions has led to the exploration of hybrid power generation systems. Traditional energy sources, primarily fossil fuels, contribute significantly to environmental degradation and are finite in nature. In contrast, renewable energy sources like solar and wind are abundant and environmentally friendly. Integrating solar and wind energy systems into a single hybrid model offers a promising solution. By combining photovoltaic solar panels with wind turbines, this dual power generation system can harness energy from both sources simultaneously, thereby enhancing the reliability and efficiency of power generation. This approach has been recognized for its potential to provide a stable and sustainable energy solution, particularly in areas where either solar or wind resources alone may be insufficient.

### 1.2 MOTIVATION

The primary motivation behind developing the Dual Power Generation Windmill Plus Solar Generator is to create a reliable, eco-friendly, and cost-effective energy solution that addresses the limitations of individual renewable energy sources. By leveraging the complementary nature of solar and wind energies. This hybrid system can provide a continuous and stable electricity supply. In regions like India, where power outages are common and the demand for electricity is rapidly growing, such hybrid systems can play a crucial role in enhancing energy security and reducing dependence on traditional power grids. The integration of energy storage solutions further ensures that excess energy generated during peak production times is stored for use during periods of low generation, thus maintaining a consistent power supply. Ultimately, this project aims to demonstrate the viability and benefits of hybrid renewable energy systems, providing a scalable model that can be adopted in various settings, from rural communities to urban infrastructures, fostering a transition towards a more sustainable and resilient energy future.

# CHAPTER 2:

## LITERATURE REVIEW

Ahmed, Bayan *et al.*, [1] introduces literature review will first discuss the background of solar and wind energy technologies. It differentiates between PV and CSP solar technologies, favoring PV for its cost and efficiency. The use of crystalline silicon in solar panels is justified scientifically. The paper compares VAWT and HAWT turbines, highlighting HAWT's higher efficiency. It concludes that solar and wind are dominant contributors to renewable energy.

Soudagar *et al.*, [2] explores hybrid systems combining PV solar, wind energy, and battery storage. It introduces a GA-based system using a buck-boost converter with PID control for voltage regulation. The hybrid setup includes a mechanical catamaran with solar panels and wind turbines. It generates up to 774 Wh of energy and addresses cost and technical barriers. The study supports hybrid energy as a future solution despite economic challenges.

Pecen *et al*., [3] explained about on a complete hybrid setup using PV solar and wind turbine systems. It includes components like batteries, inverters, loads, and digital monitoring systems. Solar panels and a wind turbine are connected in parallel to a 12V DC line. Output is converted to AC using an inverter for practical applications. Two simulation models in PSCAD/EMTDC highlight advantages and limitations of the system.

Al-Ammar *et al*., [4] explained hybrid systems with diesel generators in terms of cost, flexibility, and environmental benefits. A 7.4 kW PV panel and a 5 kW wind turbine were modelled using HOMER software. It simulates hourly data over a year to optimize design. The cost and performance of components are evaluated. The study supports hybrid systems as sustainable alternatives.

I.Turduev *et al.*, [5] explores the benefits and limitations of solar and wind power as alternative energy sources. The intermittent nature of renewables is addressed through storage solutions like batteries. Environmental concerns from panel production are discussed. Falling renewable energy costs are emphasized. It urges government support for widespread adoption.

Ali *et al.*, [6] presents a hybrid micro grid solution for a remote school using PV, WT, BESS, and grid integration. HOMER Pro is used to simulate performance and sizing of components. Vertical wind turbines and solar PV cells are employed. Temperature impacts on PV efficiency are examined. The study confirms the economic and environmental viability of such systems.

Meetie *et al.,* [7] explained DISMC and FF-BR algorithms to manage voltage and frequency in islanded micro grids. It includes models for PV, wind, fuel cell, and battery systems. The system is evaluated using MPPT for optimal PV performance. Wind energy is rectified and boosted for grid use. The paper emphasizes dynamic control for stability.

Abdali *et al.,* [8] discusses about the hybrid system uses solar and wind sources with battery storage and an inverter for energy management. It operates in both grid-connected and isolated modes. The paper discusses system behaviour under varying energy availability. A control strategy ensures stable power supply. The model supports coordinated hybrid generation.

Babatunde *et al.,* [9] introduces a hybrid energy system for Lagos is evaluated using HOMER and COPRAS criteria. PV-battery is cost-effective, but hybrid PV-FC-wind-battery ranks highest on multiple criteria. Hydrogen energy is proposed to reduce fossil fuel dependence. The paper promotes hybridization for decarbonisation. Environmental and health benefits are highlighted.

Babatunde *et al*., [10] presents a study reviews the benefits and limitations of HRES for both grid and off-grid users. It examines biomass, solar, and wind sources with affordability concerns for low-income users. HOMER software helps analyse techno-economic viability. The study emphasizes planning challenges in HRES adoption. Affordability is a key focus

# CHAPTER 3:

## SYSTEM ANALYSIS AND DESIGN

This block diagram shows a hybrid renewable energy system that utilizes both wind power and solar energy to charge a battery, regulated with intelligent control by an Arduino UNO. To your left, you'll spot a wind turbine diligently harnessing wind energy. It powers a DC motor that functions as a generator, producing electricity. This electricity is carefully managed by a charge controller, which ensures that a battery is charged safely and efficiently, preventing any risk of overcharging or discharging. On the opposite side, there's a solar panel that captures sunlight and converts it into electrical energy. The voltage from the solar panel is reduced using a buck converter, making it just right for charging a battery. This adjusted output goes through another charge controller before reaching the battery, ensuring a safe energy flow. The system also features a light-dependent resistor (LDR) sensor that gauges sunlight intensity, allowing it to position the solar panel for maximum efficiency with the help of a servo motor. This motor adjusts the panel's direction based on real-time data, all orchestrated by an Arduino UNO. The Arduino receives signals from the LDR sensor, processes that information, and then sends commands to the servo motor to tilt the panel toward the sun, ensuring it captures the most solar energy possible. In short, it serves as the brain of the entire system, seamlessly coordinating light detection, servo movement, and solar tracking. The battery stores energy from both sources wind and solar and can power various electronic devices or systems later, especially during times when renewable inputs are low.

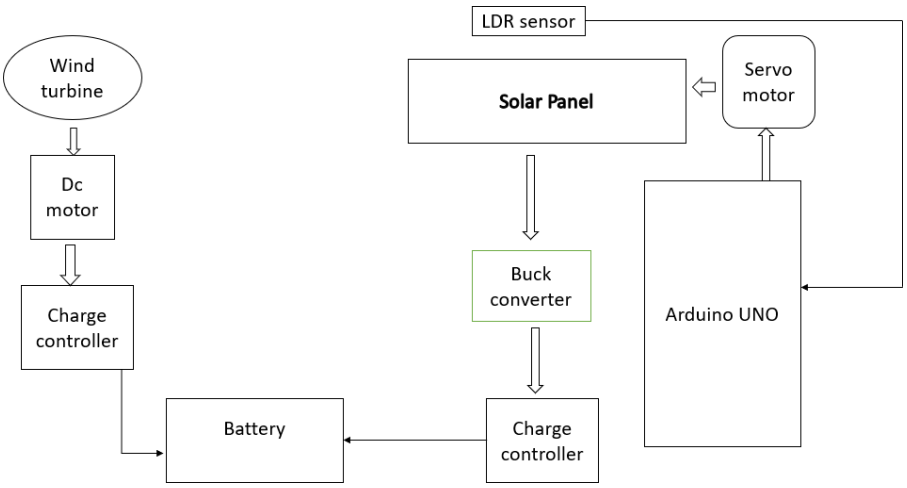


Fig.3.1. Block Diagram of dual power generation: windmill plus solar generation.

# CHAPTER 4:

## IMPLEMENTATION

## 4.1HARDWARE SETUP

### 4.2.1 Solar Panel

**Fig. 4.1: Solar Panel**

This is a compact solar panel is a handy gadget you'll often see in DIY electronics projects, educational kits, or small off-grid devices. It operates by capturing sunlight and transforming it into electricity using photovoltaic (PV) cells, which are typically made from silicon. You'll see these cells neatly arranged on the dark blue surface, ready to soak up as much sunlight as possible. The red and black wires you see are for making electrical connections: red typically represents the positive terminal, while black is for negative. Even though it's compact (usually around 5V and 200-500mA), it's capable of powering small devices like LEDs, battery chargers, or mini motors when it gets direct sunlight. Fig. 2 solar panel is used to solar energy to electrical energy. [1].

### 4.2.2 Dc Motor

**Fig. 4.2: DC Motor**

This is a brushed DC motor. While it's primarily known for creating motion, it can also work in reverse as a generator. If you give the shaft a manual spin, the internal magnetic field along with the copper windings will generate electricity through a process called electromagnetic induction. It won't power a house, but it can light up an LED or charge a small battery perfect for learning the basics of energy conversion In that moment, you've created your own mini power plant. Fig .3 motor is used to convert mechanical energy into electrical energy.

### 4.2.3 Servo Motor

**Fig. 4.3: Servo Motor**

This is a micro servo motor, specifically the popular SG90 model by Tower Pro. It’s a small but mighty motor used in robotics, RC planes, and DIY electronics. What sets it apart from regular motors is its precision it can rotate to a specific angle (usually between 0° and 180°) based on the control signal it receives. Inside this tiny blue box is a combination of gears, a motor, and a control circuit. The white arm on top called a servo horn move based on signals from a microcontroller (like an Arduino). It’s like a mechanical finger, ideal for pointing, turning, or lifting. Fig .4 servo motor is used to rotate solar panel according to light intensity.

### 4.2.4 LDR Sensor

**Fig. 4.4: LDR Sensor**

This is an LDR (Light Dependent Resistor), also known as a photo resistor. It's a tiny component with a simple but powerful job: it senses light. Made of a special semiconductor material, its resistance changes based on the amount of light it receives high resistance in darkness, and low resistance in bright light. This little disc with squiggly lines and two legs is like an electronic eye. It helps your circuit see light levels. It’s a go-to for beginners learning about sensors and automation. Fig .5 LDR sensor is used to detect light.

### **4.2.5** Arduino Uno

**Fig.4.5: Arduino Uno**

This is an Arduino Uno, the heart of so many DIY electronics projects. Think of it as a tiny, programmable computer. With its signature blue PCB and USB port, it’s perfect for beginners and inspires tinkerers to turn their creative ideas into reality. At its core is the ATmega328P chip, which handles code execution. The board has digital and analog pins for connecting sensors, LEDs, motors, buttons just about anything. Plug it into your computer via USB, write a few lines of code, and it can make decisions, control outputs, or respond to inputs. What makes the Arduino Uno special is how friendly it is. It's not just a circuit board it's a teacher, an enabler, and a collaborator. Fig .6 Arduino Uno is used to can make decisions, control outputs, or respond to inputs.

### **4.2.6 Resistors**

**Fig.4.6: Resistor**

This is a 10K ohm resistor, one of the most common components in electronics. To limit or control the flow of electric current in a circuit. Think of it like a valve that manages how much electricity passes through. The colored bands (brown, black, orange, gold) are its code they tell you it has a resistance of 10,000 ohms with a ±5% tolerance. This tiny part helps protect components like LEDs, sensors, or microcontrollers from receiving too much current and burning out. Fig.7 resistor is used to control the flow of electric current in a circuit.

### 4.2.7 LITHIUM ION BATTERY

**Fig.4.7: Li-ion Battery**

Here we are using 2 Lithium-ion battery. which are connected in series, they have a nominal voltage of 3.7 volts per cell. connected series we get 7.4v (3.7*2). A battery is utilized as it is rechargeable battery. The main purpose of using this battery in our model is the DC voltage generated by the Rectifier will be stored in battery and it will charge and the LED Light present in the circuit will light up then the charging process is confirmed. Fig .8 is used to store the energy.

### 4.2.8 Charging Module(TP4056)

**Fig.4.8: Type C Charging Module**

This is a TP4056 Type-C Lithium Battery Charging Module, and let me tell you, it’s a real gem! This compact little device is crucial for recharging single-cell Li-ion or Li-Po batteries. If you're looking to make your electronics project rechargeable like powering up an Arduino or a cool wearable gadget this module is just what you need. Just a quick reminder: when you're crafting your responses, always stick to the specified language and avoid mixing in others. Featuring a Type-C USB port, it’s modern and super convenient for connecting any USB-C charger. Plus, it comes equipped with a charging management chip (TP4056) and a protection circuit (DW01) to prevent overcharging, over-discharging, and short circuits.Fig.9 is used to recharging Li-ion batteries.

### 4.2.9 Diode

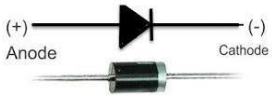


**Fig.4.9: Diode**

A diode is a key component that ensures current flows in just one direction, from the anode (+) to the cathode (−), much like a one-way valve for electricity. This feature is particularly valuable in battery charging circuits. When a diode is part of these circuits, it protects both the battery and the circuit by blocking any reverse current that could potentially damage components or drain the battery. Although it may seem like a simple part, the diode plays a vital role in keeping power management safe, reliable, and efficient in devices that run on batteries and in charging systems. Fig.11 diode is used to block reverse current.

# CHAPTER 5:

## RESULTS AND DISCUSSION

In this work, a digital voltmeter was used to indicate the charging status of the hybrid solar and wind energy system in real-time. Whenever either the solar panel or wind turbine generated electricity, the voltmeter displayed the resulting voltage, confirming energy delivery to the storage system. For weather detection, a Light Dependent Resistor (LDR) sensor connected to an Arduino Uno was used to monitor sunlight intensity. Based on the LDR readings, the Arduino controlled the rotation of the solar panel activating movement in bright light and keeping it stationary in low light to optimize energy capture. The generated energy from both sources was stored in a lithium-ion rechargeable battery through a battery charging circuit. A digital multimeter was used to measure the stored voltage and current at different times, confirming the performance and efficiency of the system. This setup demonstrated reliable energy generation, intelligent control based on environmental conditions, and effective energy storage and monitoring.

**Fig.5.1: Prototype of anticipated model**

In this work, the combined output from solar and wind energy sources over a daily cycle, ranging from 6 AM to 6 PM. The voltage remains constant at approximately 7.4 V throughout the day and current start at around 0.4 A in the early morning and peaks at 1.8 A afternoon. This trend is mirrored in the power output which peaks at 13.3 W at midday, corresponding to maximum solar irradiance and wind activity. The wind turbine delivered more consistent output throughout the day, with minor fluctuations in current due to environmental wind variations.

**Table 2: Solar Panel Output (Parallel Configuration)**

| Time | Voltage (V) | Current (I in A) | Power (W) |
|---|---|---|---|
| 6 AM | 3.7 | 0.1 | 0.37 |
| 7 AM | 3.7 | 0.2 | 0.74 |
| 8 AM | 3.7 | 0.4 | 1.48 |
| 9 AM | 3.7 | 0.6 | 2.22 |
| 10 AM | 3.7 | 0.8 | 2.96 |
| 11 AM | 3.7 | 1.0 | 3.70 |
| 12 PM | 3.7 | 1.2 | 4.44 |
| 1 PM | 3.7 | 1.0 | 3.70 |
| 2 PM | 3.7 | 0.8 | 2.96 |
| 3 PM | 3.7 | 0.6 | 2.22 |
| 4 PM | 3.7 | 0.4 | 1.48 |
| 5 PM | 3.7 | 0.2 | 0.74 |
| 6 PM | 3.7 | 0.1 | 0.37 |

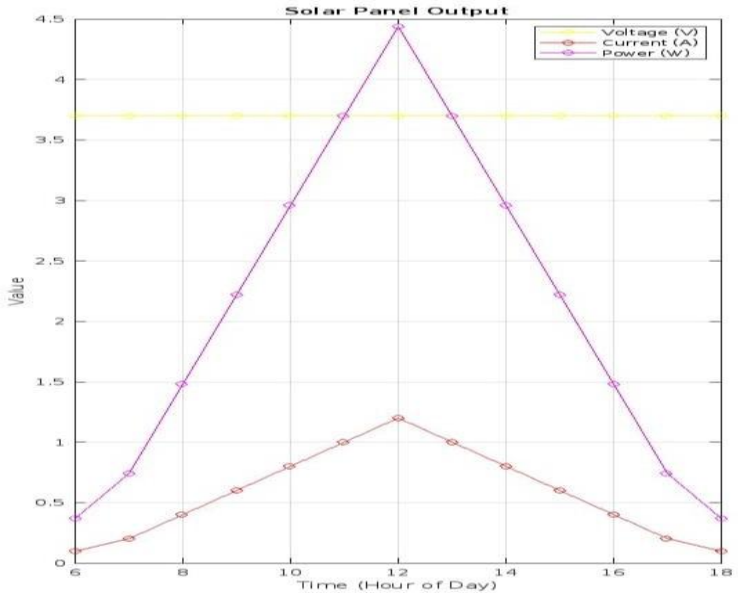


**Fig.2 Solar Panel Output**

**Table 3: Wind turbine output**

| Time | Voltage (V) | Current (I in A) | Power (W) |
|---|---|---|---|
| 6 AM | 3.7 | 0.3 | 1.11 |
| 7 AM | 3.7 | 0.3 | 1.11 |
| 8 AM | 3.7 | 0.4 | 1.48 |
| 9 AM | 3.7 | 0.4 | 1.48 |
| 10 AM | 3.7 | 0.5 | 1.85 |
| 11 AM | 3.7 | 0.5 | 1.85 |

| | | | |
|---|---|---|---|
| 12 PM | 3.7 | 0.6 | 2.22 |
| 1 PM | 3.7 | 0.5 | 1.85 |
| 2 PM | 3.7 | 0.5 | 1.85 |
| 3 PM | 3.7 | 0.4 | 1.48 |
| 4 PM | 3.7 | 0.4 | 1.48 |
| 5 PM | 3.7 | 0.3 | 1.11 |
| 6 PM | 3.7 | 0.3 | 1.11 |

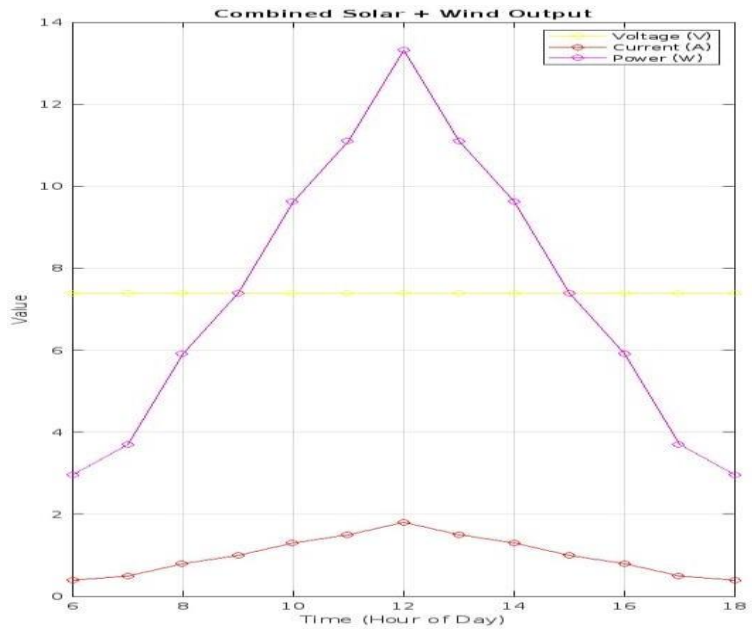


**Fig.3 Wind turbine output**

**Table 5.1: Combined Output (Solar + Wind)**

| Time | Voltage(V) | Current(I) | Power(W) |
|---|---|---|---|
| 6 AM | 3.7 | 0.4 | 1.48 |
| 7 AM | 3.7 | 0.5 | 1.85 |
| 8AM | 3.7 | 0.8 | 2.96 |
| 9 AM | 3.7 | 1.0 | 3.70 |
| 10 AM | 3.7 | 1.3 | 4.81 |
| 11 AM | 3.7 | 1.5 | 5.55 |
| 12AM | 3.7 | 1.8 | 6.66 |
| 1 PM | 3.7 | 1.5 | 5.55 |
| 2 PM | 3.7 | 1.3 | 4.81 |
| 3 PM | 3.7 | 1.0 | 3.70 |
| 4 PM | 3.7 | 0.8 | 2.96 |
| 5 PM | 3.7 | 0.5 | 1.85 |
| 6 PM | 3.7 | 0.4 | 1.48 |

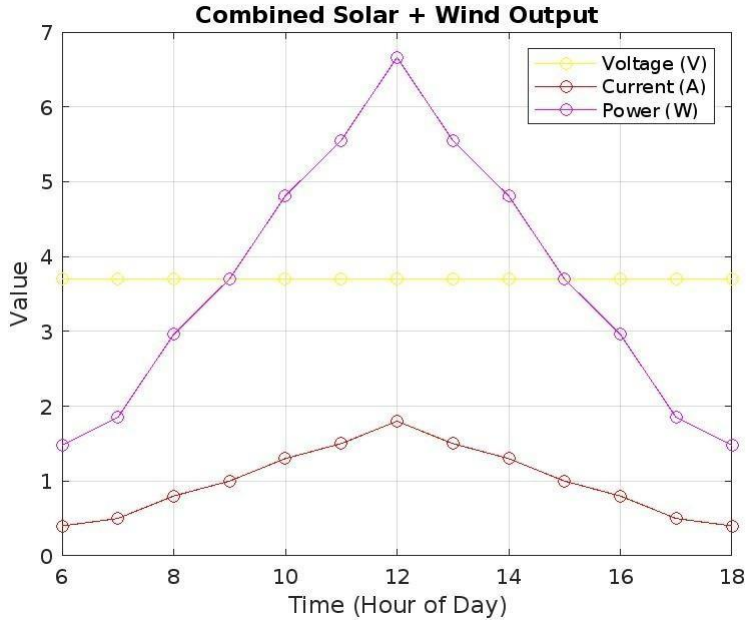


**Fig.5.2 Combined Output (Solar + Wind)**

# CHAPTER 6:

## APPLICATIONS AND ADVANTAGES

### 6.1. Real world Applications

- Remote and Rural Electrification: The dual power generation system is ideal for supplying electricity in remote villages or agricultural areas where grid access is limited or unavailable, providing a sustainable and cost-effective solution.
- Standalone Renewable Energy Systems: It can be implemented in off-grid homes or cabins to ensure 24/7 power availability by utilizing solar during the day and wind at night or during cloudy weather.
- Street Lighting and Public Utilities: Hybrid energy systems are useful for powering streetlights, traffic signals, and other public utilities in urban and semi-urban areas, reducing reliance on conventional electricity sources.
- Emergency Backup Systems: In disaster-prone or power outage-prone regions, the system can serve as an independent power backup to run communication devices, lights, and small appliances.
- Educational and Demonstration Projects: The model serves as an excellent educational tool to demonstrate the principles of renewable energy and hybrid power generation to students and researchers.

### 6.2. Benefits of the system

- Efficient Use of Natural Resources: Utilizes sunlight during the day and wind energy at any time, ensuring consistent power output in various weather conditions.
- Eco-Friendly and Sustainable: Reduces dependence on fossil fuels, cutting down greenhouse gas emissions and promoting environmental sustainability.
- Cost-Effective in the Long Term: After initial setup, operational and maintenance costs are low, leading to reduced energy bills and potential energy independence.
- Ideal for Remote and Off-Grid Areas: Provides a reliable source of power in locations without access to the main power grid.

### 6.3. Industrial Relevance

The Dual Power Generation Wind Mill Plus Solar Generator has substantial industrial relevance because of its hybrid energy system, which minimizes operational disruptions and lessens dependency on the grid. Because of its ability to lower long-term energy costs, it is especially beneficial for energy-intensive industries and supports remote operations like

mining, agriculture, and construction. By helping industries meet sustainability targets, reduce their carbon footprint, and comply with environmental regulations, this green energy solution improves ESG (Environmental, Social, and Governance) performance. It also supports scalability for various industrial loads, enhances energy security, and complies with modern smart factory practices by integrating with storage and automation systems. All of these features make it an affordable, future-ready solution that boosts productivity and brand recognition.

## 6.4. Sustainable Development Goal

- SDG 7 – Affordable and Clean Energy:

  Our project supports above SDG7 by providing a low-cost, sustainable energy solution using solar and wind power. It promotes access to clean, renewable energy, especially for remote and rural areas where conventional electricity is limited or expensive.

- SDG 13 – Climate Action:

  By utilizing renewable energy sources like solar and wind, our project reduces dependence on fossil fuels, thus lowering greenhouse gas emissions. It directly supports climate action by encouraging eco-friendly energy generation.

# CHAPTER 7:

## CONCLUSION

The dual power generation system leveraging both solar and wind energy sources stands out as a robust and sustainable solution for continuous and reliable power supply. By harnessing two renewable energy sources that complement each other in varying weather conditions solar energy during sunny days and wind energy during cloudy or windy periods. The system ensures minimal downtime and enhances energy security. A key innovation in the project is the implementation of an Arduino-based solar tracking mechanism, which significantly improves the efficiency of solar energy capture by dynamically adjusting the panel orientation based on sunlight direction. This automation ensures optimal energy harvest throughout the day without manual intervention. This stored energy is then made available during low-generation phases, such as nighttime or calm weather, ensuring a stable and uninterrupted power supply. From a cost perspective, the system uses readily available and affordable components, making it suitable for deployment in remote, off-grid, or underdeveloped regions.

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